(7)	$\neg r \vee (\neg t \vee u)$	Premises
(8)	$r \rightarrow (\neg t \vee u)$	Step (7), implication rule
(9)	r	From step (6), conjunction rule
(10)	$\neg t \vee u$	From (8), (9), disjunctive amplification
(11)	$t \rightarrow u$	step (10), implication rule
(12)	t	step (4) conjunction rule
(13)	u	step (11) & (12); Modus Ponens.

Quantifiers

A sentence that involves a variable such as x cannot be called as a statement. This cannot be called as the proposition unless the value of x is specified.

Eg:- $x+2$ is an even integer

x divides 16.

The sentences of this kinds are called open statements. If the open statement contains a variable x and when we restrict certain allowable choices for this x, this allowable choice are called the universe or the universal of discourse for open statement. Open statement containing a variable x are denoted by $p(x)$, $q(x)$, $q(x,y)$, etc.

If U is the universe for the variable x in an open statement $p(x)$ and if $a \in U$, then the proposition got by replacing x by a in $p(x)$ denoted by $p(a)$.

Eg: $p(x)$: $x+2$ is an even integer

$p(5)$: $5+2$ is an even integer (F)

$p(7)$: $7+2$ is an even integer (or T)

$q(x,y)$ is used to represent an open statement that contains 2 variables;

$q(x,y)$: "The numbers $y+2$, $x-y$ and $x+2y$ are even integers".

$q(4,2)$: "The nos 4, 2 and 8 are even integers" (T)

$q(4,1)$: "The nos 3, 3 and 6 are even integers" (F)

In the egs $p(x)$ and $q(x,y)$, some substitutions results in true statements

and other result in false statements. ∴ We can say that if

For some value of x, $p(x)$ is true ; Illy For some x, $\sim p(x)$ is False

For some x, y, $q(x,y)$ is True ; For some x, y, $\sim q(x,y)$ is False.

The phrases "For some x" & "For some x, y" are called to quantify the open statements $p(x)$ and $q(x,y)$ respectively.

There are 2 types of quantifiers.

① Existential quantifier ($\exists$) ② Universal quantifier ($\forall$)

① The phrase "for some x" are called the existential quantifier which can also be expressed as "For atleast one x" OR "There exists an x such that." etc. This quantifier is symbolically denoted as

$\exists x$: "For some x" Illy $\exists y, p(y)$, $\exists x, \exists y, q(x,y)$ or

$p(x) \rightarrow \exists x\, p(x)$ $\exists x, y, q(x,y)$

② Universal quantifier ($\forall$) are denoted by

$\forall x$, read as for all x or for any x or for each x or for every x

Eg:- $p(x)$: "The numbers $x+2$ is even integer"

The above open statement can be a quantified statement for all x, $p(x)$ $\forall x, p(x)$ is a false statement.

$r(x)$: "$2x$ is an even integer" Then the quantified statement for the above open statement $\forall x, r(x)$ is a true statement.

Note that the quantified statement there $\exists y\, r(x)$ exists an element x, $r(x)$ is also a true statement but for all $\forall x \sim r(x)$ and $\exists x \sim r(x)$ is false.

The variable x in $p(x)$ is called as free variable for $(\forall x)$

—" — x in $r(x)$ —" — bounded variable. $(\exists x)$

Quantifiers & Logical connectives

The compound open statements can be formed using logical connectives

Eg:- $\sim p(x)$; $p(x) \wedge q(x)$

suppose that the universe consists of all integers. consider the following open statement:

$p(x)$: $x \leq 3$
$q(x)$: $x+1$ is odd
$r(x)$: $x > 0$

write truth value for the following

i) $p(2)$ (ii) $\sim q(4)$ (iii) $p(-1) \wedge q(1)$
iv) $\sim p(3) \vee r(0)$ v) $p(0) \rightarrow q(0)$ T
vi) $p(1) \leftrightarrow \sim q(2)$ (vii) $p(4) \vee [q(1) \wedge r(2)]$ viii) $p(2) \wedge [q(0) \vee \sim r(2)]$

soln:
i) $p(2)$: $2 \leq 3$ (T)
ii) $\sim q(4)$: $4+1=5$ is odd (F)
iii) $p(-1) \wedge q(1)$: $(-1 \leq 3) \wedge$ ($1+1$ is odd) (F)
iv) $\sim p(3) \vee r(0)$: $\sim(3 \leq 3) \vee 0 > 0$ (F)
v) $p(0) \rightarrow q(0)$: $(0 \leq 3) \rightarrow 0+1 \rightarrow$ odd (T)
vi) ~~$\sim p(2) \vee r(1)$~~ : $\sim(3 \leq 3) \vee 0 > 0$ (T) $p(1) \leftrightarrow \sim q(2)$: $1 \leq 3 \leftrightarrow \sim$(3 odd) $\rightarrow$ F
vii) $p(4) \vee [q(1) \wedge r(2)]$: $F \vee [F \wedge T] \Rightarrow F$ (F)
viii) $p(2) \wedge [q(0) \vee \sim r(2)]$: $T \wedge [T \vee F] \Rightarrow T$

Converse, Inverse & Contrapositive for open statement

For the given open statements $p(x)$ & $q(x)$ defined for the ~~universe~~ prescribed universe or universally quantified statement,

If $\forall x (p(x) \rightarrow q(x))$ Inverse, $\forall x [\sim p(x) \rightarrow \sim q(x)]$
Converse, $\forall x [q(x) \rightarrow p(x)]$ Contrapositive, $\forall x [\sim q(x) \rightarrow \sim p(x)]$

① Let $p(x)$, $q(x)$ and $r(x)$ denote the open statements

$p(x) = x^2 - 7x + 10 = 0$
$q(x) = x^2 - 2x - 3 = 0$
$r(x) = x < 0$

Determine the truth or falsity of the following statements when the universe contains only the integers 2 & 5.

i) $\forall x [p(x) \rightarrow \sim r(x)]$ For $2 \rightarrow T$ $5 \rightarrow T$
ii) $\forall x [q(x) \rightarrow r(x)]$ For $x=2$, $(F \rightarrow \sim) = T$ ⑤ $F \rightarrow$ — T
iii) $\forall x [p(x) \rightarrow q(x)]$, F ⑤ $\rightarrow$ F

i) $\forall x [p(x) \rightarrow \sim r(x)]$

Here universe of discourse is $\{2, 5\}$

Consider $x = 2$

$\forall x [p(2) \rightarrow \sim r(2)]$

$p(x)$: $x^2 - 7x + 10 = 0$

$p(2)$: $2^2 - 7(2) + 10 = 0$

$= 4 - 14 + 10 = 0$

$p(2) = 0$ (True)

$r(x)$: $x < 0$

$r(2)$: $2 < 0$ (False)

$p(2) \rightarrow \sim r(x)$

$T \rightarrow T \Rightarrow T$

Consider $x = 5$,

$[p(5) \rightarrow \sim r(5)]$

$p(x) = x^2 - 7x + 10 = 0$

$p(5) = 25 - 35 + 10 = 0$

$p(5) = 0$ (True)

$r(x)$: $x < 0$

$r(5)$: $5 < 0$ (False)

$p(5) \rightarrow \sim r(5)$

$T \rightarrow T = $ True

$\therefore \forall x [p(x) \rightarrow \sim r(x)]$ is true

(2) If $p(x): x \geq 0$ $q(x): x^2 \geq 0$ $r(x): x^2 - 3x - 4 = 0$ $s(x) = x^2 - 3 \geq 0$

i) $\exists x [p(x) \wedge r(x)]$ ii) $\forall x [p(x) \rightarrow q(x)]$ iii) $\forall x [q(x) \rightarrow s(x)]$

Universe of discourse $[x = 0, x = 1, x = 4, x = 1.7]$

Soln:- i) $\exists x [p(x) \wedge r(x)]$ → for any 1 x

Consider $x = 0$,

$p(x) = x \geq 0$

$p(0) =$

The truth values of quantified statements "$\forall x \in S, p(x)$" is true only when $p(x)$ is true for each $x \in S$.

The statement "$\exists x \in S, p(x)$" is false only when $p(x)$ is false for $x \in S$.

Ally. for the proposition of the form "$\forall x \in S, p(x)$" to be false, it is enough to exhibit one element 'a' of 'S' such that $p(a)$ is false. So this element 'a' is called a counter example.

For the proposition of the form "$\exists x \in S, p(x)$" to be true it is enough to exhibit one element a of S such that $p(a)$ is true.

Rules of Inferences for open statement

1. universal specification
2. universal generalization

① → For $p(x)$ is an open statement for a given universe and if $\forall x p(x)$ is true, then $p(a)$ is true for each value 'a' in the universe. This is called the rule of universal specification.

For $p(x)$ is an open statement & it is true for any arbitrary x chosen from a set of S, then the statement $\forall x \in S$ the $p(x)$ is true This is known as rule of universal generalization.

Logical equivalence

The 2 quantified statements are said to be logically equivalent whenever they have same truth value in all the possible situations.

i) $\forall x [p(x) \wedge q(x)] \Leftrightarrow [\forall x, p(x) \wedge \forall x, q(x)]$

ii) $\exists x [p(x) \wedge q(x)] \Leftrightarrow [\exists x\ p(x) \wedge \exists x\ q(x)]$

iii) $\exists x [p(x) \rightarrow q(x)] \Leftrightarrow [\exists x \sim p(x) \vee \exists x\ q(x)]$

Rules of negation of a quantified statement;

To construct the negation of the quantified statement, change of the quantifier from the universal to existential and vice versa & also replace the open statement by its negation.

$\sim\{\forall x, p(x)\} \equiv \exists x \{\sim p(x)\}$

$\sim\{\exists x, p(x)\} \equiv \forall x \{\sim p(x)\}$

Negate & simplify the open statements.

1) $\exists x [p(x) \vee q(x)] \longrightarrow$ (negate) $\forall x [\sim p(x) \wedge \sim q(x)]$

2) $\forall x [p(x) \wedge \sim q(x)] \longrightarrow \exists x [\sim p(x) \vee q(x)]$

3) $\forall x [p(x) \to q(x)] \longrightarrow \exists x [p(x) \wedge \sim q(x)]$

4) $\exists x [\{p(x) \vee q(x)\} \to r(x)] \longrightarrow \forall x [\{p(x) \vee q(x)\} \wedge \sim r(x)]$

Check the validy of the following argument

1) For the universe of all the people, consider a open statement,

$m(x)$: X is a mathematics professor

$c(x)$: X has studied calculus.

Argument : All mathematics professors have studied calculus.

Leo is a mathematics professor.

$\therefore$ Leo has studied calculus.

$\forall x [m(x) \to c(x)]$

$m(a)$

$\therefore c(a)$

Steps	Reasons
1. $\forall x [m(x) \to c(x)]$	Premises
2. $m(a)$	//
3. $m(a) \to c(a)$	Step 1 using universal specification
4. $\therefore c(a)$	From step 2, 3 by modus ponens

2) Consider the universe of all triangles in the plane. In conjunction with the open statement

$p(t)$: t has two sides of equal length

$q(t)$: t is an isosceles triangle.

$r(t)$: t has two angles of equal measure

Argument : In a triangle xyz, there is not pair of angles of equal measure.

If a triangle has 2 sides of equal length, then it is isosceles.

If a Δ is isosceles, then it has 2 angles of ~~each~~ equal measure

$\therefore \Delta XYZ$ has no 2 sides of equal length.

Symbolic form

$\exists x\, Nr(a)$

$\forall_l (p(l) \to q(l))$

$\forall_l (q(l) \to r(l))$

$\therefore \exists p(a)$

Steps	Reasons
① $\exists x\, Nr(a)$	Premises
② $\forall_l (p(l) \to q(l))$	Premises
③ $p(a) \to q(a)$	step ② by universal specification
④ $\forall_l (q(l) \to r(l))$	Premises
⑤ $q(a) \to r(a)$	step ④ by universal specification
⑥ $p(a) \to r(a)$	steps ③ & ⑤, rule of syllogism
⑦ $\therefore \exists p(a)$	steps ① and ⑥, rule of modus tollens

3) check the validy of following argument. Consider the universe to be the set of all quadrilateral and the open statements are

$p(x)$: x is a square $q(x)$: x has 4 sides

The arguments are: All squares have 4 sides

The quadrilateral ABCD has 4 sides

$\therefore$ ABCD is a square.

Symbolic form

$\forall_x [p(x) \to q(x)]$

$q(a)$

$\therefore p(a)$

steps	Reasons
① $\forall_x [p(x) \to q(x)]$	Premise
② $p(a) \to q(a)$	step ① using universal specification
③ $q(a)$	Premise

Since $p(a) \to q(a)$ is true and $q(a)$ is True. Therefore $p(a)$ must be False. Hence the given statement is invalid.

4) Find whether the argument is valid or not. Consider all the engineering students as universe. Arguments:

No engineering students of 1st or 2nd semester studies logic design.

Anil is an engg. student who studies Logic design

$\therefore$ Anil is not in 2nd semester.

Open strat :- $p(x)$: x is in first semester $r(x)$: x studies logic design

$q(x)$: x is in 2nd sem.

$\forall x [p(x) \wedge q(x) \rightarrow \sim r(x)]$

$r(a)$

$\therefore \sim q(a)$

Steps	Reasons
① $\forall x[p(x) \wedge q(x) \rightarrow \sim r(x)]$	Premises
② $p(a) \wedge q(a) \rightarrow \sim r(a)$	step ①, universal specification
③ $r(a)$	Premises
④ $r(a) \rightarrow \sim[p(a) \wedge q(a)]$	step ②, contrapositive
⑤ $\sim[p(a) \wedge q(a)]$	step ③ and ④ using modus Ponens
⑥ $\sim p(a) \vee \sim q(a)$	step ⑤ by Demorgan's Law
⑦ $\sim q(a)$	step ⑥ conjunctive simplication

The argument is valid.

Rule of universal generalization

The rule states that if an open statement, $p(x)$ is proved to be true when x is replaced by arbitrarily chosen element 'c' from the universe Then the universally quantified statement $\forall x\, p(x)$ is also true.

Eg:- Let $p(x)$, $q(x)$ and $r(x)$ be an open statement. S.T. the following arguments are valid.

$\forall x [p(x) \rightarrow q(x)]$

$\forall x [q(x) \rightarrow r(x)]$

$\therefore \forall x [p(x) \rightarrow r(x)]$

Steps	Reasons
① $\forall x[p(x) \rightarrow q(x)]$	Premise
② $p(c) \rightarrow q(c)$	step ①, universal specification
③ $\forall x[q(x) \rightarrow r(x)]$	Premise
④ $q(c) \rightarrow r(c)$	step ③, universal specification
⑤ $p(c) \rightarrow r(c)$	step ② & ④, rule of syllogism
⑥ $\forall x[p(x) \rightarrow r(x)]$	step ⑤ using rule of generalization

1) Find whether the following is a valid argument for which the universe is a set of all students. Arg are

No engg student is bad in studies.

Anil is not bad in studies.

∴ Anil is an engineering students.

Symbolic form :

Given that: $q(x)$: x is bad in studies.

$p(x)$: x is an engineering student.

$\forall x [p(x) \rightarrow \sim q(x)]$

$\sim q(a)$

$\therefore p(a)$

Steps	Reasons
① $\forall x [p(x) \rightarrow \sim q(x)]$	Premises
② $p(a) \rightarrow \sim q(a)$	step ①, universal specification
③ $\sim q(a)$	Premise
④ $\therefore p(a)$	step ② & ③, modus ponens.

Not valid.

2) $\forall x [p(x) \rightarrow \{q(x) \wedge r(x)\}]$

$\forall x [p(x) \wedge s(x)]$

$\therefore \forall x [r(x) \wedge s(x)]$

UNIT-2 RELATIONS

Cartesian product or product set or cross product (×)

If A and B are 2 non-empty sets, we define the product set or Cartesian product is a set of all ordered pairs (a,b) where $a \in A, b \in B$

$\therefore A \times B = \{(a,b) \mid a \in A \text{ \& } b \in B\}$ Note: $A \times B \neq B \times A$

Eg:- If $A = \{1,2,3\}$ and $B = \{r,s\}$

$A \times B = \{(1,r), (1,s), (2,r), (2,s), (3,r), (3,s)\}$

$B \times A = \{(r,1), (r,2), (r,3), (s,1), (s,2), (s,3)\}$

Relation:

Let A & B be the two non-empty sets, a relation R from A to B is a subset of $A \times B$. $R \subseteq (A \times B)$ and $(a,b) \in R$

we can say that a is related to b by R i.e., aRb.

If a is not related to b by R, then represented as $a \not R b$.

If a & b are equal, then aRa or $R \subseteq (A \times A)$

Eg:- $A = \{1,2,3\}$ and $B = \{r,s\}$

$R = \{(1,r), (2,s), (3,r)\}$

If A and B are a set of real nos., then we define relation R from A to B as, aRb iff $a=b$.

① Let $A = \{1,2,3,4,5,6\}$

Define the relation R Less than A when A related to B [iff $a<b$]

$R = \{(1,2), (1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,4), (3,5), (4,5)\}$

② Let $A = \{a,b,c\}$ and $B = \{1,2,3\}$

i) Is $R = \{(a,1), (b,2), (c,2)\}$ a relation from A to B. Yes

ii) Is $Q = \{(1,a), (2,b)\}$ a relation from A to B. No (B to A)

iii) Is $P = \{(a,a), (b,c), (b,a)\}$ a relation from $A \times A$. Yes

③ Let $A = \{1,2,3,4\}$; $R = \{(a,b) \mid a \text{ divides } b\}$

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

Properties / Types of Relation

① Reflexive relation

A relation R on set A is said to be reflexive, if ~~for~~ ~~if~~

$\forall a \in A, (a,a) \in R$.

$(a,b,c) \times (a,b,c)$

	a	b	c
a	aa	ab	ac
b	ba	bb	bc
c	ca	cb	cc

check

1) $R = \phi$ ×
2) $R = A \times A$ ✓
3) $R = \{(a,a), (b,b), (c,c)\}$ ✓
4) $R = \{(a,a), (b,b), (c,c), (a,b), (b,c)\}$ ✓
5) $R = \{(a,b), (b,a), (a,a), (b,b)\}$ × (no(c,
6) $R = \{(a,b), (b,c), (a,c)\}$ ×

② Irreflexive relation

A relation R on set A is said to be reflexive if $\forall a \in A, (a,a) \notin R$

③ Symmetric relation

A relation R on set A is said to be symmetric if $\forall a,b \in A$,

~~if~~ $(a,b) \in R$ & $(b,a) \in R$

Eg:- 1. $R = \{(a,b), (b,a)\}$ ✓
2. $R = \{(b,c), (c,b), (b,b), (c,c)\}$ ✓
3. $R = \{(a,a), (b,b), (c,c)\}$ ✓
4. $R = \phi$ ×
5. $R = A \times A$ ✓
6. $R = \{(a,b), (b,c), (a,c)\}$ ×
7. $R = \{(a,b), (b,a), (a,c)\}$ × no((c,a)

④ Asymmetric

A relation R on set A is said to be asymmetric $\forall (a,b \in A$,

$(a,b) \in R$ & $(b,a) \notin R$.

Eg:- $R = \{(a,b), (b,c), (c,a)\}$ ✓
$R = \{(a,c), (b,c), (c,b)\}$ ✓
$R = \{(a,a), (b,b), (c,c)\}$ ×
$R = \{(a,b), (b,a), (a,a), (b,b)\}$ ×
$R = A \times A$ ×
$R = \{(a,b), (a,c), (b,c)\}$ ✓

⑤ ~~Cofactor~~ Antisymmetric relation

A relation R on set A is said to be antisymmetric if

if $(a,b) \in R$ but if $a \neq b$

$\forall a,b \in A$ $(a,b) \in R$ & $(b,a) \in R$ if $a = b$. Diagonal ele must be present

Eg:- $R = \{(a,b), (b,c), (a,c)\}$ ✓
$R = \{(a,b), (a,a), (b,b)\}$ ✓
$R = \{(a,a), (b,b), (c,c)\}$ ✓
$R = \{(a,b), (b,a), (b,c), (c,c)\}$ ×
$R = A \times A$ ✓
$R = \{(a,b), (b,c), (a,c), (c,a), (a,a), (c,c)\}$ ×

Transitive

A relation R on set A is transitive if $\forall a, b \in A$, $(a,b) \in R$ $(b,c) \in R$ then $(a,c) \in R$.

① $\{(a,b)(b,a)(a,a),(b,b)\}$ ✓

② $\{(a,a)(b,b)(c,c)\}$ ✓

③ $\{(a,b)\}$ ✓

④ $\{(a,b),(a,c)\}$ ✓

⑤ $\{(a,b),(c,b)\}$ ✓

⑥ $\{(a,b),(b,c),(a,c),(a,a)\}$ ✓

⑦ $\{(c,a)(b,c)\}$ ✓

$A = \{1,2,3\}$

Determine

i) $R_1 = \{(1,2),(2,1),(1,3),(3,1)\}$ → symm & Irreflexive

$R_2 = \{(1,1),(2,2),(3,3),(2,3)\}$ → Reflexive, transitive

$R_3 = \{(1,1),(2,2),(3,3)\}$ → Reflexive, symmetric, transitive

$R_4 = \{(1,1),(2,2),(3,3),(2,3),(3,2)\}$ → Reflexive, transitive

$R_5 = \{(1,1),(2,3),(3,3)\}$ → Reflexive, Transitive

$R_6 = \{(2,3),(3,4),(2,4)\}$ → Transitive

$R_7 = \{(1,3),(3,2)\}$ → Irreflexive

Matrix of a Relation $[M_R]$ or $[MR]$

Consider a set $A = [A_1, A_2, A_3, \ldots A_m]$ m and n → rows & columns

and $B = [B_1, B_2, B_3, \ldots B_n]$ of order $m \times n$. respectively.

And R is a relation from A to B, then R is represented by the matrix

$m \times n$ $M_R = [M_{ij}]$

where $M_{ij} = \begin{cases} 1 & \text{iff } (a_i, b_j) \in R \\ 0 & \text{iff } (a_i, b_j) \notin R \end{cases}$

Thus $M \times N$ matrix formed by $M_{i,j}$ is called relation matrix for R and it is denoted by $[MR]$. Since M_R ~~only~~ contains only 0s and 1s it is also called as zero-one matrix for a relation.

Eg:- ① $A = \{1,2,3\}$; $B = \{r,s\}$

$R = \{(1,r),(2,s),(3,r)\}$

$$M_R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

② $$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$ For the given matrix (3×4) find R.

$A = \{a_1, a_2, a_3\}$ $B = \{b_1, b_2, b_3, b_4\}$

$R = \{(a_1,b_1),(a_1,b_4),(a_2,b_2),(a_2,b_3),(a_3,b_1),(a_3,b_3)\}$

Digraph: Let R be a relation on a finite set A, then R can be represented pictorically.

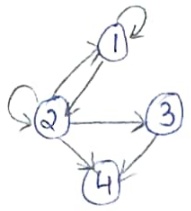
Draw a small circle or bullet for each element of A and label the circle with corresponding element of A. This circles or the bullets are called vertices or nodes. Draw an arrow called the edge from a vertex x to a vertex y iff $(x,y) \in R$. The resulting pictorical representation of R is called a directed graph or digraph of a relation.

The no. of edges (arrows) terminating at a vertex is called the indegree of a vertex.

The no. of " leaving a vertex is called outdegree of a vertex.

Eg:- $A = \{1,2,3,4\}$

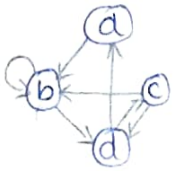
$R = \{(1,1), (1,2), (2,2), (2,3), (2,4), (3,4)\}$



	1	2	3	4
Indegree	3	2	1	2
Outdegree	2	4	1	1

① $A = \{a,b,c,d\}$

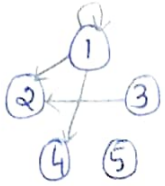
$R = \{(a,b), (b,b), (b,d), (c,b), (c,d), (d,a), (d,c)\}$



	a	b	c	d
Indegree	1	3	1	2
Outdegree	1	2	2	2

② $A = \{1,2,3,4,5\}$

$R = \{(1,1), (1,2), (1,4), (3,2)\}$



	1	2	3	4	5
indegree	1	2	0	1	0
outdegree	3	0	1	0	0

③ $A = \{1,2\}$ $B = \{p,q,r,s\}$; $R = \{(1,q), (1,r), (2,p), (2,q), (2,s)\}$

Write matrix of relation.

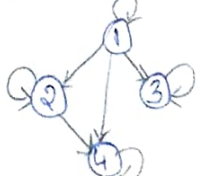
$$M_R = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

④ Let $A = \{1,2,3,4\}$. If R be the relation on A defined by x is related to y iff x divides y.

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

i) R → set of ordered pairs → ~~$R = \{(1,1), (2,1), (2,2), (3,3), (3,1), (4,1), (4)\}$~~

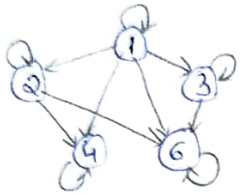
ii) Draw digraph of the relation & determine the indegree & outdegree of the vertices in the digraph.



	1	2	3	4
Indegree	1	2	2	3
Outdegree	4	2	1	1

4) Consider $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by a is related to b iff a is a multiple of b. Represent the relation R as a matrix and draw its digraph.

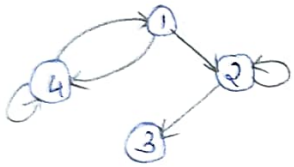
$R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (6,6)\}$



$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

5) Find the relation represented by a digraph given below. Also write down the matrix.

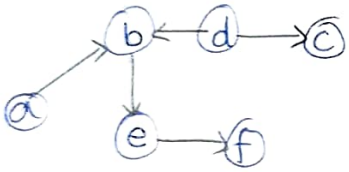
i)



$R = \{(1,2), (1,4), (2,2), (2,3), (4,4), (4,1)\}$

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

ii)



$R = \{(a,b), (b,e), (d,b), (d,c), (e,f)\}$

$$M_R = \begin{bmatrix} a & b & c & d & e & f \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Equivalence relation

A relation R on set A is said to be an equivalence relation on A if R is reflexive and R is transitive on A and R is symmetric.

Eg: ① Let $A = \{1,2,3,4\}$ and

$R = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,3), (3,3), (4,4)\}$

$\Rightarrow$ $(1,1), (2,2), (3,3), (4,4) \in R$

$(a,a) \in R$ $\therefore$ R is a reflexive relation

$\Rightarrow$ $(1,2), (2,1) \in R$ $(3,4), (4,3) \in R$

$(a,b), (b,a) \in R$ $\therefore$ R is a symmetric relation

$\Rightarrow$ $(1,2), (2,1), (1,1) \in R$ & $(2,1), (1,2) (2,2) \in R$

$(4,3), (3,4), (3,3) \in R$ $\therefore$ R is transitive on A.

$\therefore$ R is an equivalence relation.

Let $A = \{1,2,3,4\}$

$R = \{(1,1),(1,2),(2,1),(2,2),(3,1),(3,3),(1,3),(4,1),(4,4)\}$ be a relation

Determine whether R is a equivalence relation.

Here $(1,1),(2,2),(3,3),(4,4) \in R \rightarrow$ reflexive

$(1,3) \in R$ but $(3,1) \notin R \rightarrow$ Not symmetric

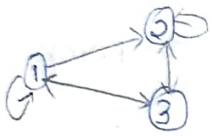
$\therefore$ Not equivalence.

(5) $A = \{a,b,c\}$ Soln: $R = \{(a,a),(a,c),(b,b),(c,c)\}$

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Not symmetric no (c,a)

(6)



$R = \{(1,1),(1,2),(1,3),(2,2),(2,3),(3,1),(3,2)\}$

$(3,3) \notin R$ not reflexive

Not equivalence

Equivalence classes

Let R be an equivalence relation on set A and $a \in A$, then the set of all those elements x (i.e. (x,a)) which are related to a by R are called equivalence class.

$R(a)$ or $\bar{a}$ or $[a] = \{x \in A \mid (x,a) \in R\}$

Eg: ① Let $A = \{1,2,3\}$

$R = \{(1,1),(1,3),(2,2),(3,1),(3,3)\}$

$[1] = \{1,3\}$; $[2] = \{2\}$; $[3] = \{1,3\}$

② Let $A = \{1,2,3,4,5,6\}$

$R = \{(1,1),(1,3),(1,5),(3,1),(3,3),(3,5),(5,1),(5,3),(5,5),(2,2),(2,6),(6,2),(6,6),(4,4)\}$

$[1] = \{1,3,5\}$ $[4] = \{4\}$

$[2] = \{2,6\}$ $[5] = \{1,3,5\}$

$[3] = \{1,3,5\}$ $[6] = \{2,6\}$

③ Let $A = \{1,2,3,4,5\}$

$R = \{(1,1),(2,2),(3,3),(4,4),(5,5),(1,2),(2,1),(4,5),(5,4)\}$

$[1] = \{1,2\}$ $[4] = \{4,5\}$

$[2] = \{2,1\}$ $[5] = \{5,4\}$

$[3] = \{3\}$ $[6] = \{\}$

Partition of a set

Let A be a non-empty set. Suppose there exists a non-empty subsets $A_1, A_2, A_3, \ldots A_k$ of A such that the following 2 conditions hold i) A is the union of $A_1, A_2, A_3, \ldots A_k$

$$A = A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_k$$

ii) Any two of the subsets $A_1, A_2, A_3 \ldots A_k$ are disjoints i.e,

$A_i \cap A_j = \phi$ for $i \neq j$. Then the set be

$P = \{A_1, A_2, A_3, \ldots A_k\}$ is called a partition of A and also $A_1, A_2, A_3, \ldots A_k$ are called the blocks or the cells of the partition.

1) For The equivalence relation $R = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,3), (3,3), (4,4)\}$ defined on set $A = \{1,2,3,4\}$. Determine partition of set A. & equivalence classes.

Soln:- $[1] = \{1,2\}$ $[3] = \{3,4\}$

$[2] = \{1,2\}$ $[4] = \{3,4\}$

~~(If This equiva)~~ The equivalence relatn of this equivalence class

[1] and [3] are distinct. or [1] and [4] [2] and [3]

$P = [[1] \cup [3]] = [\{1,2\} \cup \{3,4\}]$

$A = [[1] \cup [3]] = [1,2,3,4]$

2) Consider a set $A = \{1,2,3,4,5\}$ and equivalence relation

$R = \{(1,1), (2,2), (2,3), (3,2), (3,3), (4,4), (4,5), (5,4), (5,5)\}$ defined on A

Find the partition of A induced on R.

$[1] = \{1\}$ $[3] = \{2,3\}$ $[5] = \{4,5\}$

$[2] = \{2,3\}$ $[4] = \{4,5\}$

$P = [[1] \cup [2] \cup [4]] = [\{1\} \cup \{2,3\} \cup \{4,5\}] = [1,2,3,4,5] = A$

3) Let $A = \{1,2,3,4,5,6,7\}$

$A = \{1,2\} \cup \{3\} \cup \{4,5,7\} \cup \{6\}$ Find the relation.

$R = \{(1,1), (1,2) \rightarrow 1R1, 1R2, 2R2, 2R1$ Since we have $\{1,2\}$ in block 1

$\rightarrow$ 3R3 (2nd block)

since in the 3rd block we have {4,5,7}

4R4, 4R5, 4R7, 5R5, 5R4, 5R7, 7R7, 7R4, 7R5

in 4th block

6R6

R = {(1,1), (1,2), (2,2), (2,1), (4,4), (4,5), (4,7), (5,5), (5,4), (5,7), (7,7), (7,4), (7,5), (6,6)}

4) Consider A = {a,b,c,d,e} & P = {{a,b}, {c,d}, {e}} be partition.

find equivalence relation using the partition.

aRa, aRb, bRa, bRb, cRd,

~~Partial~~ Partial orders